



Import History XML Connector File Specification

Data Import Guide

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Revision History

Revision	Date	Description
2.0	4/8/24	Added file naming convention and recommendation in the “INTERFACE REPORT FILE NAMING CONVENTION” section.
1.0	3/1/24	WFM R8.0 document. • Under the section “Mapping to NICE WFM Data Requirements”, added ContactsActive to Queue Data. Also added to “Input File Sample” and “Sample XML Report” sections. • In the “Sample DTD File Format for XML Interface Protocol”, corrected the element “Backlog” from using a lower-case l to upper-case “BackLog”. (WFM-138521) • Under the section “Mapping to the NICE WFM Data Requirements”, in the Agent - Queue Data tables, changed the HandledTime element name to HandleTime.

Notice

This document is for information purposes and describes certain software, features, and capabilities of NICE products. Every effort has been made to ensure the information is correct and current. However, information is subject to change without notice.

Table of Contents

Revision History	3
Overview	5
Scope of this Guide	5
Customer Responsibilities	5
Initial Setup	6
Types of Media Channels	6
Mapping to NICE WFM Data Requirements	6
Queue Data	7
Agent Queue Data	13
Agent System Data	16
Agent Activity Data	18
XML File – Interface Protocol	18
Directory Path (Location) on WFM Server for Delivery of XML File	18
Contents and Timing of Interface Reports	19
Interface Report File Naming Convention	19
Method of Retrieving Missing Interval Data	19
XML Reports Contacts and Rules	20
Input File Sample	21
Sample DTD (Document Type Definition) File Format for XML Interface Protocol	21
Sample XML Report	22
Validating the XML Report	24

Overview

If you are running NICE WFM R8.x, the Import History XML Connector allows customers to import Contact Type (CT) and Agent historical data items from a XML file.

The Import History feature for the Import History XML Connector can be used:

- By new customers to reduce the data entry time required to populate WFM historical statistics during the initial database build.
- As an automatic data posting method to populate data from a contact routing system that routes work items (calls, emails, chats, back office tickets, etc.) to agents.

For the purpose of this document, the term *Vendor system* may be used to describe the customer's system/module that contains the statistics pertaining to work items.

Scope of this Guide

This guide covers the following main topics:

- The initial setup required before processing the import history file.
- A description of the types of media channels for NICE WFM.
- Mapping to NICE WFM Queue, Agent-Queue, Agent System, and Agent Activity data requirements.
- The Import History XML connector file interface protocol.
- Import History XML connector sample DTD (Document Type Definition).
- Sample Import History XML report.
- Validating the Import History XML report.

Note: This guide does not include implementation details to automate the import of CT and agent historical data.

Customer Responsibilities

The customer may submit a request to the NICE Professional Services group to configure the Import History program to run automatically at pre-determined intervals. The customer may also request training on how to invoke the program manually.

The customer creates the input file of historical data according to the specifications in this document. For the Import History program to locate and process the file, the customer must provide a “pathway” between the WFM server and the file. The NICE Professional Services group will work with you to define the file path.

Data is imported into the QUEUESTATS, AGTQUEUESTATS, AGTSYSSTATS, and ACTIVITYLOGDTL tables.

Initial Setup

Prior to processing an import file, you must define:

- a CT for each type of contact needed for forecasting and scheduling
- at least one Queue for each CT
 - each ACD Queue Tag must match a Contact Queue Identifier in the import file

An import file may not exceed 5 MB in size. If the input file is greater than 5MB, the user must segment the file into smaller files.

Types of Media Channels

The NICE WFM system supports all media channels (media types) used within contact centers. However, within NICE WFM, media channels are distinguished in four categories; *Immediate Response*, *Deferrable Response*, *Outbound Campaign Management (OCM)*, and *Outbound*.

Media Type	Description
<i>Immediate Response</i> type Contacts	Contacts requiring immediate or near immediate response by agents (employees). (i.e., Inbound Voice, Chat, Text (SMS), Web Response)
<i>Deferrable Response</i> type Contacts	Contacts requiring a deferrable response by agents (employees). (i.e., Email, Fax, BackOffice Work Item)
<i>Outbound Campaign Management</i> type Contacts	Contacts delivered to agents (employees) from an Outbound Dialer type system. (i.e., Outbound Dialer, Predictive Dialer) Note: This functionality requires the enablement of the OCM (Outbound Campaign Management) feature within NICE WFM.
<i>Outbound</i> type Contacts	Outbound Contacts delivered to agents (employees) in which NICE WFM uses a time series method for forecasting

Mapping to NICE WFM Data Requirements

The historical database tables within NICE WFM are designed to store statistics in four different groupings:

- **Queue Data:** (noted as *QueueData* within the XML report) Contact activity routed through (or pegged against) specific queues within the *Vendor system*.
- **Agent-Queue Data:** (noted as *AgentQueueData* within the XML report) Agent statistics for each queue in which they exercised (handled) activity.
- **Agent System Data:** (noted as *AgentSystemData* within the XML report) Agent statistics that commonly cannot be reported by queue (Login time, Ready time, Not Ready time are examples).
- **Agent Activity Data:** (noted as *AgentActivityData* within the XML report) Agent activities such as Login events, Logout events, and Not Ready events (with associated reason codes if applicable).

QUEUE DATA

Following is a list of data elements required to meet NICE WFM data collection requirements for the Queue level of statistics.

Interval: Summary data representing 15- or 30-minute interval

The detailed descriptions of each data item follow:

QueueValue Contact Queue Identifier _[key] - 1 to 100 ASCII characters	This represents the Contact Router Contact Queue identifier.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required – key field – must be unique
	<i>Deferrable Response</i>	Required – key field – must be unique
	<i>Outbound Campaign</i>	Required – key field – must be unique
	<i>Outbound</i>	Required – key field – must be unique

ContactsReceived [count]	The number of Contacts that entered the Contact Router system during this interval on this Contact Queue.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Required
	<i>Deferrable Response</i>	Required
	<i>Outbound Campaign</i>	Not Required
	<i>Outbound</i>	Not Required

AbandonedShort [count]	Also known as <i>Abandoned Before Grade of Service</i> . This is the number of contacts abandoned prior to a specific time threshold which the customer or vendor views as “short abandoned”. The specific time threshold is determined by the Contact Router system. If the vendor system supports the Abandoned Contacts metric but does not differentiate AbandonedShort from AbandonedLong, this field may be omitted from the report.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required (if available)
	<i>Deferrable Response</i>	Not Required (but may be supplied)
	<i>Outbound Campaign</i>	Not Required (ignored by WFM) For Queues defined within WFM with <i>Outbound Campaign Management</i> Media Type, the WFM interface will reference the Var2, for Outbound OCM CTs.
	<i>Outbound</i>	Not Required (ignored by WFM) For Queues defined within WFM with <i>Outbound</i> Media Type, the WFM interface will reference the Var2, for Outbound CTs.

AbandonedLong [count]	Also known as <i>Abandoned After Grade of Service</i> . This is the number of contacts abandoned after a specific time threshold was met. The specific time threshold is determined by the Contact Router system. If the vendor system supports the Abandoned Contacts metric but does not differentiate AbandonedShort from AbandonedLong, the <i>total</i> Abandoned Contacts count may be supplied in this field.
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	Media Type	Data Element
	<i>Immediate Response</i>	Required (if available, otherwise omitted)
	<i>Deferrable Response</i>	Not Required (but may be supplied)
	<i>Outbound Campaign</i>	Not Required (ignored by WFM)
	<i>Outbound</i>	Not Required (ignored by WFM)

<u>HandledShort</u> [count]	The number of contacts handled (serviced) before a particular grade of service time was met. The actual grade of service time threshold is determined by the Contact Router system. If the vendor system supports the Handled Contacts metric but does not differentiate HandledShort from HandledLong, this field may be omitted from the report.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required (if available, otherwise omitted)
	<i>Deferrable Response</i>	Not Required (but may be supplied)
	<i>Outbound Campaign</i>	Not Required For Queues defined within WFM with <i>Outbound Campaign Management</i> Media Type, the WFM interface will reference the Var1, for Outbound OCM CTs.
	<i>Outbound</i>	Not Required For Queues defined within WFM with <i>Outbound</i> Media Type, the WFM interface will reference the Var1, for Outbound CTs.

<u>HandledLong</u> [count]	For <i>Immediate Response</i> and <i>Deferrable Response</i> type contacts, this is the number of handled contacts handled (serviced) after a particular grade of service time was met. The actual grade of service time threshold is determined by the Contact Router system. If the vendor system supports the Handled Contacts metric but does not differentiate HandledShort from HandledLong, the <i>total</i> Handled Contacts count may be supplied in this field. Not applicable to <i>Outbound Campaign Management</i> type contacts.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required
	<i>Deferrable Response</i>	Required
	<i>Outbound Campaign</i>	Not Applicable For Queues defined within WFM with <i>Outbound Campaign Management</i> Media Type, the WFM interface will reference the RPC (Right Party Contacts) and WPC (Wrong Party Contacts) values to derive this value.
	<i>Outbound</i>	Not Applicable

<u>HandleTime</u> [duration reported in seconds]	For <i>Immediate Response</i> and <i>Deferrable Response</i> type contacts, this is the total handle time (plus Hold Time, if applicable) of all handled contacts during this interval on this Contact Queue. Not applicable to <i>Outbound Campaign Management</i> type contacts.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required

	<i>Deferrable Response</i>	Required
	<i>Outbound Campaign</i>	Not Applicable For Queues defined within WFM with <i>Outbound Campaign Management</i> Media Type, the WFM interface will reference the RPC Handle Time and WPC Handle Time to derive this value.
	<i>Outbound</i>	Not Applicable

<u>HoldTime</u> [duration reported in seconds]	The total Hold Time (the amount of time during this interval the customer was placed on Hold) for Handled, RPC, and WPC contacts that occurred during this interval on this Contact Queue.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required (if available, otherwise omitted)
	<i>Deferrable Response</i>	Not Required
	<i>Outbound Campaign</i>	Required (if available, otherwise omitted)
	<i>Outbound</i>	Required (if available, otherwise omitted)

<u>WorkTime</u> [duration reported in seconds]	The total After Contact Work (or WrapUp) Time for all handled contacts that occurred during this interval on this Contact Queue. Note: The Work Time is being applied to the Queue of the previously handled contact.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required (if available, otherwise omitted)
	<i>Deferrable Response</i>	Required (if available, otherwise omitted)
	<i>Outbound Campaign</i>	Required (if available, otherwise omitted)
	<i>Outbound</i>	Required (if available, otherwise omitted)

<u>QueueDelayTime</u> [duration reported in seconds]	The total amount of time customers wait in queue before the contact is presented to an agent for handling. For inbound voice, this time does not include IVR time.	
	For WFM to accurately calculate Average Speed of Answer (ASA), this value should be pegged to the same interval in which the contact is pegged as <i>Handled</i> .	
	Media Type	Data Element
	<i>Immediate Response</i>	Required (if available, otherwise omitted)
	<i>Deferrable Response</i>	Required (if available, otherwise omitted)
	<i>Outbound Campaign</i>	Not Required (ignored by WFM)
	<i>Outbound</i>	Not Required (ignored by WFM)

<u>SvcLvlPct</u> <u>(Service Level Percent)</u> [reported as a percentage to two decimals]	For <i>Immediate Response</i> type contacts, WFM will use HandledShort, HandledLong, AbandonedShort, and AbandonedLong to derive the Service Level Percent value, therefore this field will be ignored.	
	For <i>Deferrable Response</i> type contacts, this may reflect the same derivation as Immediate Response type contacts (as described directly above) or it <i>may</i> equate to the percentage of contacts the agent responded to the customer on or before the service level threshold.	

	The actual Service Level Threshold (time value) is determined per Contact Queue by the Contact Router system.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Required (ignored by WFM)
	<i>Deferrable Response</i>	Required (if available, otherwise omitted)
	<i>Outbound Campaign</i>	Not Applicable
	<i>Outbound</i>	Not Applicable

BackLog [count]	For <i>Deferrable Response</i> type contacts, this is the total number of contacts received in previous intervals for the Contact Queue that have not been worked / completed / handled / abandoned. Not applicable to <i>Immediate Response</i> type Contacts or <i>Outbound Campaign Management</i> type Contacts. Note: <u>BackLog</u> = <u>BackLogNotExpired</u> + <u>BackLogExpired</u> For <i>Deferrable Response</i> type contacts, if the vendor supports the BackLog metric concept, any two of the three backlog data elements may be provided. If the vendor supports the BackLog metric concept but does not differentiate between Expired and Not Expired, the total BackLog value may be supplied in this BackLog field.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Required (if available, otherwise omitted)
	<i>Outbound Campaign</i>	Not Applicable
	<i>Outbound</i>	Not Applicable

BackLogNotExpired [count]	For <i>Deferrable Response</i> type contacts, this is the total number of contacts received in previous intervals for the Contact Queue that have not been worked / completed / handled / abandoned and have not yet “aged” past a specified time limit. The service threshold has <i>not</i> passed for these contacts. These contacts may still be worked in future periods and the service level objective may still be achieved. Not applicable to <i>Immediate Response</i> type contacts or <i>Outbound Campaign Management</i> type contacts. Note: <u>BackLog</u> = <u>BackLogNotExpired</u> + <u>BackLogExpired</u> For <i>Deferrable Response</i> type contacts, if the vendor supports the BackLog metric concept, any two of the three backlog data elements may be provided. If the vendor supports the BackLog metric concept but does not differentiate between Expired and Not Expired, the total BackLog value should be supplied in the BackLog field.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Required (if available, otherwise omitted)
	<i>Outbound Campaign</i>	Not Applicable
	<i>Outbound</i>	Not Applicable

<u>BackLogExpired</u> [count]	For <i>Deferrable Response</i> type contacts, this is the total number of contacts received in previous intervals for the Contact Queue that have not been worked / completed / handled / abandoned before the designated acceptable time limit. The service threshold has passed for these contacts. These contacts may still be worked in future periods but the service level objective will <i>not</i> be achieved. Not applicable to <i>Immediate Response</i> type contacts or <i>Outbound Campaign Management</i> type contacts. Note: <u>BackLog</u> = <u>BackLogNotExpired</u> + <u>BackLogExpired</u> For <i>Deferrable Response</i> type contacts, if the vendor supports the BackLog metric concept, any two of the three backlog data elements may be provided. If the vendor supports the BackLog metric concept but does not differentiate between Expired and Not Expired, the total BackLog value may be supplied in the BackLog field.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Required (if available, otherwise omitted)
	<i>Outbound Campaign</i>	Not Applicable
	<i>Outbound</i>	Not Applicable

<u>RightPartyContacts</u> [count]	For <i>Outbound Campaign Management</i> type contacts, this is the number of (Outdialer) Right Party Contacts that occurred during this interval on this Outbound Contact Queue. If the vendor system does not differentiate RPC from WPC, the total count of Handled Contacts for outdialer calls should be supplied to this field. Not applicable to <i>Immediate Response</i> type contacts or <i>Deferrable Response</i> type contacts.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Not Applicable
	<i>Outbound Campaign</i>	Required
	<i>Outbound</i>	Required

<u>RightPartyTalkTime</u> <u>(RPC Handle Time)</u> [duration reported in seconds]	For <i>Outbound Campaign Management</i> type contacts, this is the handle time (plus Hold Time, if applicable) of (Outdialer) Right Party Contacts that occurred during this interval on this Outbound Contact Queue. If the vendor system does not differentiate RPC from WPC, the total Handle Time of Handled Contacts for outdialer calls should be supplied to this field. Not applicable to <i>Immediate Response</i> type contacts or <i>Deferrable Response</i> type contacts. Note: The data item names RightPartyTalkTime and RPC Handle Time are synonymous.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Not Applicable
	<i>Outbound Campaign</i>	Required
	<i>Outbound</i>	Required

<u>WrongPartyContacts</u> [count]	For <i>Outbound Campaign Management</i> type contacts, this is the number of (Outdialer) Wrong Party Contacts that occurred during this interval on this Outbound Contact Queue. If the vendor system does not differentiate RPC from WPC, the total count of Handled Contacts for outdialer calls should be supplied to the RPC Contacts field and this field should be omitted. Not applicable to <i>Immediate Response</i> type contacts or <i>Deferrable Response</i> type contacts.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Not Applicable
	<i>Outbound Campaign</i>	Required (if available, otherwise omitted)
	<i>Outbound</i>	Required (if available, otherwise omitted)

<u>WrongPartyTalkTime</u> <u>(WPC Handle Time)</u> [duration reported in seconds]	For <i>Outbound Campaign Management</i> type contacts, this is the handle time (plus Hold Time if applicable) of (Outdialer) Wrong Party Contacts that occurred during this interval on this Outbound Contact Queue. If the vendor system does not differentiate RPC from WPC, the total Handle Time of Handled Contacts for outdialer calls should be supplied to the RPC Handle Time field and this field should be omitted. Not applicable to <i>Immediate Response</i> type contacts or <i>Deferrable Response</i> type contacts. Note: The data item names WrongPartyTalkTime and WPC Handle Time are synonymous.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Not Applicable
	<i>Outbound Campaign</i>	Required (if available, otherwise omitted)
	<i>Outbound</i>	Required (if available, otherwise omitted)

<u>ContactsActive</u> [count]	The number of Contacts worked in the interval but answered in a previous interval.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required
	<i>Deferrable Response</i>	Required
	<i>Outbound Campaign</i>	Not Applicable
	<i>Outbound</i>	Required

AGENT QUEUE DATA

Following is a list of data elements required to meet NICE WFM data collection requirements for the Agent-Queue level of statistics. **Note:** Some data elements are only applicable to certain types of media.

Interval: Summary data representing 15- or 30-minute interval

The detailed descriptions of each data item follow:

<u>QueueValue</u> Contact Queue Identifier [key] - 1 to 100 ASCII characters	This represents the Contact Router Contact Queue identifier.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required – key field – must be unique
	<i>Deferrable Response</i>	Required – key field – must be unique
	<i>Outbound Campaign</i>	Required – key field – must be unique
	<i>Outbound</i>	Required – key field – must be unique

<u>AgentValue</u> Agent Identifier [key] - 1 to 256 ASCII characters	This represents the Contact Router agent identifier. Commonly, this is equivalent to an agent Logon ID, Extension, Name, Email Address, etc.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required – key field – must be unique
	<i>Deferrable Response</i>	Required – key field – must be unique
	<i>Outbound Campaign</i>	Required – key field – must be unique
	<i>Outbound</i>	Required – key field – must be unique

<u>Handled</u> [count]	The total number of <i>Immediate Response</i> and <i>Deferrable Response</i> type contacts handled (serviced) by this agent during this interval on this Contact Queue.	
	Not applicable to <i>Outbound Campaign Management</i> type contacts.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required
	<i>Deferrable Response</i>	Required
	<i>Outbound Campaign</i>	Not Required For Queues defined within WFM with <i>Outbound Campaign Management</i> Media Type, the WFM interface will reference the RPC (Right Party Contacts) and WPC (Wrong Party Contacts) values to derive this value.
	<i>Outbound</i>	Not Required

<u>HandleTime</u> [duration reported in seconds]	For <i>Immediate Response</i> and <i>Deferrable Response</i> type contacts, this is the total handle time (plus Hold Time, if applicable) of all contacts handled by this agent during this interval on this Contact Queue. Not applicable to <i>Outbound Campaign Management</i> type contacts.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required
	<i>Deferrable Response</i>	Required
	<i>Outbound Campaign</i>	Not Required For Queues defined within WFM with <i>Outbound Campaign Management</i> Media Type, the WFM interface will reference the RPC Handle Time and WPC Handle Time to derive this value.
	<i>Outbound</i>	Not Required

<u>HoldTime</u> [duration reported in seconds]	The total Hold Time (the amount of time during this interval the customer was placed on Hold) for Handled, RPC, and WPC contacts that occurred for this agent during this interval on this Contact Queue.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required if used in this environment, or may be omitted.
	<i>Deferrable Response</i>	Not Required
	<i>Outbound Campaign</i>	Required (if available, otherwise omitted)
	<i>Outbound</i>	Required (if available, otherwise omitted)

<u>WorkTime</u> [duration reported in seconds]	This is the total After Contact Work (or WrapUp) Time for all handled contacts for this agent that occurred during this interval on this Contact Queue. Note: The Work Time is being applied to the Queue of the previously handled contact.	
	Media Type	Data Element
	<i>Immediate Response</i>	Required (if available, otherwise omitted)
	<i>Deferrable Response</i>	Required (if available, otherwise omitted)
	<i>Outbound Campaign</i>	Required (if available, otherwise omitted)
	<i>Outbound</i>	Required (if available, otherwise omitted)

<u>RightPartyContacts</u> [count]	For <i>Outbound</i> and <i>Outbound Campaign Management</i> type contacts, this is the number of (Outdialer) Right Party Contacts that occurred for this agent during this interval on this Outbound Contact Queue. If the vendor system does not differentiate RPC from WPC, the total count of Outdialer Handled Contacts for this agent should be supplied to this field. Not applicable to <i>Immediate Response</i> type contacts or <i>Deferrable Response</i> type contacts.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Not Applicable
	<i>Outbound Campaign</i>	Required
	<i>Outbound</i>	Required

<u>RightPartyTalkTime (RPC Handle Time)</u> [duration reported in seconds]	For <i>Outbound</i> and <i>Outbound Campaign Management</i> type contacts, this is the handle time (plus Hold Time, if applicable) of (Outdialer) Right Party Contacts that occurred for this agent during this interval on this Outbound Contact Queue. If the vendor system does not differentiate RPC from WPC, the total Handle Time of Outdialer Handled Contacts for this agent should be supplied to this field. Not applicable to <i>Immediate Response</i> type contacts or <i>Deferrable Response</i> type contacts. Note: The data item names RightPartyTalkTime and RPC Handle Time are synonymous.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Not Applicable
	<i>Outbound Campaign</i>	Required
	<i>Outbound</i>	Required

<u>WrongPartyContacts</u> [count]	For <i>Outbound</i> and <i>Outbound Campaign Management</i> type contacts, this is the number of (Outdialer) Wrong Party Contacts that occurred for this agent during this interval on this Outbound Contact Queue. If the vendor system does not differentiate RPC from WPC, the total count of Outdialer Handled Contacts for this agent should be supplied to the RPC Contacts field and this field should be omitted. Not applicable to <i>Immediate Response</i> type contacts or <i>Deferrable Response</i> type contacts.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Not Applicable
	<i>Outbound Campaign</i>	Required (if available, otherwise omitted)
	<i>Outbound</i>	Required (if available, otherwise omitted)

<u>WrongPartyTalkTime</u> <u>(WPC Handle Time)</u> [duration reported in seconds]	For <i>Outbound</i> and <i>Outbound Campaign Management</i> type contacts, this is the handle time (plus Hold Time, if applicable) of (Outdialer) Wrong Party Contacts that occurred for this agent during this interval on this Outbound Contact Queue. If the vendor system does not differentiate RPC from WPC, the total Handle Time of Outdialer Handled Contacts for this agent should be supplied to the RPC Handle Time field and this field should be omitted.	
	Not applicable to <i>Immediate Response</i> type contacts or <i>Deferrable Response</i> type contacts.	
	Note: The data item names WrongPartyTalkTime and WPC Handle Time are synonymous.	
	Media Type	Data Element
	<i>Immediate Response</i>	Not Applicable
	<i>Deferrable Response</i>	Not Applicable
	<i>Outbound Campaign</i>	Required (if available, otherwise omitted)
	<i>Outbound</i>	Required (if available, otherwise omitted)

AGENT SYSTEM DATA

Following is a list of data elements required to meet WFM data collection requirements for the Agent System level of statistics.

Note: Statistics collected from the Agent System report are stored by various groupings within the NICE WFM system; Agent reporting, Management Unit reporting (grouping of agents), and Contact Type reporting (grouping of queues). For some of these statistics to be grouped by queue, the NICE WFM interface application must proportionally distribute these data items to the queues based on the amount of TalkTime and WorkTime the agent had on each queue during the interval.

Interval: Summary data representing 15- or 30-minute interval

The detailed descriptions of each data item follow:

<u>AgentValue</u> Agent Identifier [key] - 1 to 256 ASCII characters	This represents the Contact Router agent identifier. Commonly, this is equivalent to an agent Logon ID, Extension, Name, Email Address, etc.	
	Media Type	Data Element
	<i>All media types</i>	Required – key field – must be unique

<u>InternalContacts</u> [count]	The number of Internal Contacts (agent-to-agent, or agent-to-supervisor type contacts) received by this Agent during this interval. These may be internal voice calls, internal chats, internal emails, etc.	
	Note: These are pegged to the receiving agent and not the initiating agent.	
	Media Type	Data Element
	<i>All media types</i>	Required (if available, otherwise omitted)

<u>InternalHandleTime</u> [duration reported in seconds]	The Handle Time of Internal Contacts received by this Agent that occurred during this interval.	
	Media Type	Data Element
	<i>All media types</i>	Required (if available, otherwise omitted)

<u>ReadyTime</u> [duration reported in seconds]	The amount of time during this interval the Agent spent in the <i>Ready</i> state (available and ready to receive contacts). This value should not exceed the interval length for a particular agent.	
	Media Type	Data Element
	<i>All media types</i>	Required (if available, otherwise omitted)

<u>NotReadyTime</u> [duration reported in seconds]	The amount of time during this interval the Agent spent in a <i>Not Ready</i> state. <i>Not Ready</i> is the time an agent is logged in, not available to receive contacts, and not in a work related state. It can possibly be used for personal time. Note: This value should not include any After Contact Work activity. This value should not exceed the interval length for a particular agent.	
	Media Type	Data Element
	<i>All media types</i>	Required (if available, otherwise omitted)

<u>OutboundContacts</u> [count]	The number of outbound contacts made by this Agent during this interval. Note: This value represents <i>agent-initiated-out</i> contacts whereby an agent makes an outbound contact directly from their desktop. This value does not represent the Outbound Campaign Management type of contacts.	
	Media Type	Data Element
	<i>All media types</i>	Required (if available, otherwise omitted)

<u>OutboundHandleTime</u> [duration reported in seconds]	The Handle Time of Outbound Contacts made by this Agent that occurred during this interval. Note: This value represents <i>agent-initiated-out</i> contacts whereby an agent makes an outbound contact directly from their desktop. This value does not represent the Outbound Campaign Management type of contacts.	
	Media Type	Data Element
	<i>All media types</i>	Required (if available, otherwise omitted)

<u>LoginTime</u> [duration reported in seconds]	The amount of time the Agent was logged in during this interval. This value should not exceed the interval length for a particular interval.	
	Media Type	Data Element
	<i>All media types</i>	Required (if available, otherwise omitted)

AGENT ACTIVITY DATA

Following is a list of data elements required to meet NICE WFM data collection requirements for the Agent Activity level of statistics.

Interval: Summary data representing a 24-hour interval

Note: *Unlike the other three reports described in this document, the Agent Activity report consists of data representing a 24-hour interval.*

The detailed descriptions of each data item follow:

AgentValue Agent Identifier _ [key] - 1 to 256 ASCII characters	This represents the Contact Router agent identifier. Commonly, this is equivalent to an agent Logon ID, Extension, Name, Email Address, etc.	
	Media Type	Data Element
	<i>All media types</i>	Required – key field – must be unique

code [#PCDATA] – Parsed Character Data (mixed content)	The agent activity which represents a logon, logoff, available, or Not Ready (Away) state.	
	Media Type	Data Element
	<i>All media types</i>	Required if Agent Activity Data is being reported.

DateTime [#PCDATA] – Parsed Character Data (mixed content)	The date and time the agent entered the state, in the format CCYYMMDDThhmm.	
	Media Type	Data Element
	<i>All media types</i>	Required if Agent Activity Data is being reported.

reason [#PCDATA] – Parsed Character Data (mixed content)	The agent activity which represents a logon, logoff, available, or Not Ready (Away) state.	
	Used in conjunction with the <i>code</i> data element (if the <i>code</i> value reflects a Not Ready state).	
	Media Type	Data Element
	<i>All media types</i>	Optional (if available, otherwise omitted)

XML File – Interface Protocol

The data received is formatted in XML (eXtensible Markup Language).

DIRECTORY PATH (LOCATION) ON WFM SERVER FOR DELIVERY OF XML FILE

The WFM server directory path where the XML files should be delivered is:

/totalview/ftp/switches/customer<X>/<acdid#>/

where:

- customer<X> is the NICE WFM database (customer) partition number
- <acdid#> is the ACD Link Identification Number for this particular interface link, as defined within the NICE WFM system.

For example, a directory path for a customer partition of number 1 and ACD ID of 21 would appear as follows:

/totalview/ftp/switches/customer1/21/

The Customer ID and ACD ID are items that may be communicated to the customer at the time of WFM server deployment.

Most commonly, the XML file is presented (delivered) to the WFM server by File Transfer Protocol (FTP), or by a configured Network Drive Map.

CONTENTS AND TIMING OF INTERFACE REPORTS

The data within the XML files sent by the *Vendor system* are summary statistics representing queue and agent activity that occurred during the previous interval, and are transmitted shortly after the most recent interval has ended. The NICE WFM server is configured to receive these reports 24 hours a day, 7 days a week. If no data is available for a particular interval, no data records are present within the XML reports.

INTERFACE REPORT FILE NAMING CONVENTION

The XML files transmitted to the WFM server must adhere to a predefined file naming convention established during interface design. This naming convention must be consistently applied for each reporting interval. To ensure uniqueness per interval, the same file name should be used each time data is collected for that interval, even if the data is updated multiple times.

It is recommended that the interval time be incorporated into the file name using the format **MMDDYY.HHMM.xml**. For example, for data corresponding to the April 6th interval at 12:00 PM, the file should be named **040625.1200.xml**, regardless of the number of updates made during that interval.

For instance, if one XML file is presented to the WFM server that contains all segments of historical data (Queue, Agent-Queue, Agent System, and the optional Agent Activity), a file format of **MMDDYY.HHMM.xml** might be used, since it represents a date/time stamp of which the data (reports) within the file represent. Likewise, if four individual XML files are presented to the WFM server, each interval whereby one contains Queue data, another Agent-Queue data, another Agent System data, and the optional Agent Activity data, the four reports might have names like **Q_MMDDYY.HHMM.xml**, **AQ_MMDDYY.HHMM.xml**, **AS_MMDDYY.HHMM.xml** and **AA_MMDDYY.HHMM.xml**.

In the examples above, MMDDYY.HHMM would appear as 112514.1330 representing the reporting interval span between 01:30PM and 02:00PM on November 25, 2014.

METHOD OF RETRIEVING MISSING INTERVAL DATA

The NICE WFM interface application continually searches the above mentioned directory for files to process. Any files located that match the file naming convention are processed.

XML REPORTS CONTACTS AND RULES

- WFM 8.x supports either 15- or 30-minute interval granularities for the Queue, Agent-Queue, and Agent System reports. This means the data within these reports must be summarized to represent activity that occurred during either a 15- or 30-minute period. The determination of whether 15- or 30-minute summarization is used is based on a WFM setting determined at the time of WFM server deployment. For 30-minute interval environments, the reporting interval must begin on the hour or half-hour. For 15-minute interval environments, the reporting interval must begin on the hour, quarter-hour, half-hour, or three-quarter-hour.
- Unlike the Queue, Agent-Queue, and Agent System reports, the Agent Activity Report is a 24-hour interval summary report, which (if used) may be included in the XML file once per day, generally shortly after midnight. The Agent Activity Report may be submitted within its own XML file, if the sender chooses.

Note:

1. For current data posting and reposting, no more than one interval should be included in each XML file, with filename = start of interval (the same convention used for other ACD interfaces).
2. To repost multiple interval data (Bulk Import), create a single XML file which contains all segments of historical data (Queue, Agent-Queue, Agent System) for all intervals in this single file. The naming convention will remain the same 'MMDDYY.HHMM.xml' with HHMM signifying the starting first interval from all those intervals.

- While submitting a single XML file that contains a single interval of statistics for processing, the following rules apply:
 - The XML file must contain properly formatted nodes for all 15/30 minute reports; Queue, Agent-Queue, and Agent System. At a minimum, the XML must contain nodes as shown directly below. A processing error will occur if the XML file is missing any of these nodes as indicated.

```
<HistPlugin>
  <DataSourceNode>
    <Vendor></Vendor>
    <QueueNode>
      <TimePeriod><DateTime>20140102T0900</DateTime></TimePeriod>
    </QueueNode>
  </DataSourceNode>
  <DataSourceNode>
    <Vendor></Vendor>
    <AgentQueueNode>
      <TimePeriod><DateTime>20140102T0900</DateTime></TimePeriod>
    </AgentQueueNode>
  </DataSourceNode>
  <DataSourceNode>
    <Vendor></Vendor>
    <AgentSystemNode>
      <TimePeriod><DateTime>20140102T0900</DateTime></TimePeriod>
    </AgentSystemNode>
  </DataSourceNode>
</HistPlugin>
```

- While submitting multiple XML files, each of which contain a single report for processing, the following rules apply:
 - Each file is processed, however, all reports must be received within 48 hours for the interval to post within NICE WFM. If all reports are not received within 48 hours, the entire interval will be discarded.

- The NICE WFM system is configured such that, by default, no interval may be reposted to the WFM database. If an interval of statistics is received for which records already exist within the WFM database, the redundant interval reports will be discarded. In this scenario, if the XML file contains multiple intervals, the redundant interval is skipped and processing continues to the next interval in the XML file.

Input File Sample

SAMPLE DTD (DOCUMENT TYPE DEFINITION) FILE FORMAT FOR XML INTERFACE PROTOCOL

```

<!ELEMENT duration (totalseconds)>
<!ELEMENT totalseconds (#PCDATA)>
<!ELEMENT count (#PCDATA)>
<!ELEMENT percentage (#PCDATA)>
<!ELEMENT average (#PCDATA)>
<!ELEMENT OutputType (#PCDATA)>
<!ELEMENT TimePeriod (DateTime,period?)>
<!ELEMENT DateTime (#PCDATA)>
<!ELEMENT period (#PCDATA)>
<!ELEMENT HistPlugin (DataSourceNode*)>
<!-- A DataSourceNode is a set of data that came from a single contributor plug-in. This plug-in delivers data from an
ACD system. All data elements provided in a node pertain to the statistics period provided in that node. -->
<!ELEMENT DataSourceNode
(Vendor, (QueueNode*|AgentQueueNode*|AgentSystemNode*|AgentActivityNode*))>
<!ELEMENT Vendor (#PCDATA)>
<!ELEMENT QueueNode (TimePeriod,QueueData*)>
<!ELEMENT AgentQueueNode (TimePeriod,AgentQueueData*)>
<!ELEMENT AgentSystemNode (TimePeriod,AgentSystemData*)>
<!ELEMENT AgentActivityNode (TimePeriod,AgentActivityData*)>
<!ELEMENT QueueData
(QueueValue,ContactsReceived?,AbandonedShort?,AbandonedLong?,HandledShort?,Ha
ndledLong?,HandleTime?,HoldTime?,WorkTime?,QueueDelayTime?,SvcLvlPct?,BackLog
?,BackLogExpired?,BackLogNotExpired?,RightPartyContacts?,RightPartyTalkTime?,
WrongPartyContacts?,WrongPartyTalkTime?,ContactsActive?)>
  <!ELEMENT QueueValue (#PCDATA)>
  <!ELEMENT ContactsReceived (count)>
  <!ELEMENT AbandonedShort (count)>
  <!ELEMENT AbandonedLong (count)>
  <!ELEMENT HandledShort (count)>
  <!ELEMENT HandledLong (count)>
  <!ELEMENT HandleTime (duration)>

```

```

<!ELEMENT HoldTime (duration)>
<!ELEMENT WorkTime (duration)>
<!ELEMENT QueueDelayTime (duration)>
<!ELEMENT SvcLvlPct (percentage)>
<!ELEMENT BackLog (count)>
<!ELEMENT BackLogExpired (count)>
<!ELEMENT BackLogNotExpired (count)>
<!ELEMENT RightPartyContacts (count)>
<!ELEMENT RightPartyTalkTime (duration)>
<!ELEMENT WrongPartyContacts (count)>
<!ELEMENT WrongPartyTalkTime (duration)>
<!ELEMENT ContactsActive (count)>

<!ELEMENTAgentQueueData
  (QueueValue,AgentValue,Handled?,HandleTime?,HoldTime?,WorkTime?,RightPartyCon
  tacts?,RightPartyTalkTime?,WrongPartyContacts?,WrongPartyTalkTime?)>
  <!ELEMENT AgentValue (#PCDATA)>
  <!ELEMENT Handled (count)>

<!ELEMENTAgentSystemData
  (AgentValue,InternalContacts?,InternalHandleTime?,ReadyTime?,NotReadyTime?,Ou
  tboundContacts?,OutboundHandleTime?,LoginTime?)>
  <!ELEMENT InternalContacts (count)>
  <!ELEMENT InternalHandleTime (duration)>
  <!ELEMENT ReadyTime (duration)>
  <!ELEMENT NotReadyTime (duration)>
  <!ELEMENT OutboundContacts (count)>
  <!ELEMENT OutboundHandleTime (duration)>
  <!ELEMENT LoginTime (duration)>

<!ELEMENT AgentActivityData (AgentValue,Event*)>
  <!ELEMENT Event (code,DateTime,reason?)>
    <!ELEMENT code (#PCDATA)>
    <!ELEMENT reason (#PCDATA)>

```

Sample XML Report

```

<?xml version="1.0"?><!DOCTYPE HistPlugin SYSTEM
"../../../../../../../../../../../../swxiface/plugins/com.iex.tv.swxiface/src/main/resources/conf/plugi
n/HistPlugin.dtd">
<HistPlugin>
  <DataSourceNode>
    <Vendor></Vendor>
    <QueueNode>
      <TimePeriod><DateTime>20140102T0900</DateTime></TimePeriod>
      <QueueData>
        <QueueValue>44</QueueValue>                                <!-- Simulation (Erlang) based CT -->
        <AbandonedShort><count>5</count></AbandonedShort>
        <AbandonedLong><count>2</count></AbandonedLong>
        <HandledShort><count>6</count></HandledShort>
        <HandledLong><count>3</count></HandledLong>
        <HandleTime><duration><totalseconds>200</totalseconds></duration></HandleTime>
        <HoldTime><duration><totalseconds>50</totalseconds></duration></HoldTime>
        <WorkTime><duration><totalseconds>100</totalseconds></duration></WorkTime>
        <ContactsActive><count>3</count></ContactsActive>

      <QueueDelayTime><duration><totalseconds>125</totalseconds></duration></QueueDelayTime>
    </QueueData>
  </QueueData>
  <QueueValue>45</QueueValue>                                <!-- Workload CT -->

```

```

        <ContactsReceived><count>99</count></ContactsReceived>
        <AbandonedShort><count>5</count></AbandonedShort>
        <AbandonedLong><count>2</count></AbandonedLong>
        <HandledShort><count>6</count></HandledShort>
        <HandledLong><count>3</count></HandledLong>
        <HandleTime><duration><totalseconds>200</totalseconds></duration></HandleTime>
        <HoldTime><duration><totalseconds>50</totalseconds></duration></HoldTime>
        <WorkTime><duration><totalseconds>100</totalseconds></duration></WorkTime>
        <ContactsActive><count>1</count></ContactsActive>

    <QueueDelayTime><duration><totalseconds>125</totalseconds></duration></QueueDelayTime>
        <SvcLvlPct><percentage>99.99</percentage></SvcLvlPct>
        <BackLogExpired><count>2</count></BackLogExpired>
        <BackLogNotExpired><count>3</count></BackLogNotExpired>
    </QueueData>
</QueueNode>
</DataSourceNode>
<DataSourceNode>
    <Vendor></Vendor>
    <AgentQueueNode>
        <TimePeriod><DateTime>20140102T0900</DateTime></TimePeriod>
        <AgentQueueData>
            <QueueValue>44</QueueValue>
            <AgentValue>30128</AgentValue>
            <Handled><count>10</count></Handled>
            <HandleTime><duration><totalseconds>120</totalseconds></duration></HandleTime>
            <HoldTime><duration><totalseconds>50</totalseconds></duration></HoldTime>
            <WorkTime><duration><totalseconds>80</totalseconds></duration></WorkTime>
        </AgentQueueData>
        <AgentQueueData>
            <QueueValue>45</QueueValue>
            <AgentValue>30128</AgentValue>
            <Handled><count>20</count></Handled>
            <HandleTime><duration><totalseconds>350</totalseconds></duration></HandleTime>
            <HoldTime><duration><totalseconds>55</totalseconds></duration></HoldTime>
            <WorkTime><duration><totalseconds>50</totalseconds></duration></WorkTime>
        </AgentQueueData>
        <AgentQueueData>
            <QueueValue>409</QueueValue>
            <AgentValue>40258</AgentValue>
            <Handled><count>0</count></Handled>
            <HandleTime><duration><totalseconds>0</totalseconds></duration></HandleTime>
            <HoldTime><duration><totalseconds>0</totalseconds></duration></HoldTime>
            <WorkTime><duration><totalseconds>0</totalseconds></duration></WorkTime>
        </AgentQueueData>
    </AgentQueueNode>
</DataSourceNode>
<DataSourceNode>
    <Vendor></Vendor>
    <AgentSystemNode>
        <TimePeriod><DateTime>20140102T0900</DateTime></TimePeriod>
        <AgentSystemData>
            <AgentValue>30128</AgentValue>
            <InternalContacts><count>1</count></InternalContacts>

        <InternalHandleTime><duration><totalseconds>101</totalseconds></duration></InternalHandleTime>
            <ReadyTime><duration><totalseconds>300</totalseconds></duration></ReadyTime>
            <NotReadyTime><duration><totalseconds>90</totalseconds></duration></NotReadyTime>
            <OutboundContacts><count>2</count></OutboundContacts>

        <OutboundHandleTime><duration><totalseconds>200</totalseconds></duration></OutboundHandleTime>
            <LoginTime><duration><totalseconds>900</totalseconds></duration></LoginTime>
        </AgentSystemData>
        <AgentSystemData>
            <AgentValue>30143</AgentValue>
            <InternalContacts><count>2</count></InternalContacts>

        <InternalHandleTime><duration><totalseconds>102</totalseconds></duration></InternalHandleTime>
            <ReadyTime><duration><totalseconds>300</totalseconds></duration></ReadyTime>
            <NotReadyTime><duration><totalseconds>90</totalseconds></duration></NotReadyTime>

```

```

        <OutboundContacts><count>2</count></OutboundContacts>

    <OutboundHandleTime><duration><totalseconds>200</totalseconds></duration></OutboundHandleTime>
        <LoginTime><duration><totalseconds>900</totalseconds></duration></LoginTime>
    </AgentSystemData>
    <AgentSystemData>
        <AgentValue>40258</AgentValue>
        <InternalContacts><count>3</count></InternalContacts>

    <InternalHandleTime><duration><totalseconds>103</totalseconds></duration></InternalHandleTime>
        <ReadyTime><duration><totalseconds>300</totalseconds></duration></ReadyTime>
        <NotReadyTime><duration><totalseconds>90</totalseconds></duration></NotReadyTime>
        <OutboundContacts><count>2</count></OutboundContacts>

    <OutboundHandleTime><duration><totalseconds>200</totalseconds></duration></OutboundHandleTime>
        <LoginTime><duration><totalseconds>900</totalseconds></duration></LoginTime>
    </AgentSystemData>
</AgentSystemNode>
</DataSourceNode>
<DataSourceNode>
    <Vendor></Vendor>
    <AgentActivityNode>
        <TimePeriod><DateTime>20080101T0000</DateTime></TimePeriod>
        <AgentActivityData>
            <AgentValue>30128</AgentValue>
            <Event>
                <code>Login</code>
                <DateTime>20080101T0700</DateTime>
            </Event>
            <Event>
                <code>Logout</code>
                <DateTime>20080101T1000</DateTime>
                <reason>1</reason>
            </Event>
            <Event>
                <code>Login</code>
                <DateTime>20080101T1100</DateTime>
            </Event>
            <Event>
                <code>Logout</code>
                <DateTime>20080101T1500</DateTime>
                <reason>2</reason>
            </Event>
        </AgentActivityData>
        <AgentActivityData>
            <AgentValue>30143</AgentValue>
            <Event>
                <code>Login</code>
                <DateTime>20080101T1030</DateTime>
            </Event>
            <Event>
                <code>Logout</code>
                <DateTime>20080101T1130</DateTime>
                <reason>2</reason>
            </Event>
        </AgentActivityData>
    </AgentActivityNode>
</DataSourceNode>
</HistPlugin>

```

Validating the XML Report

Follow these steps to validate your XML file against HistPlugin.dtd.

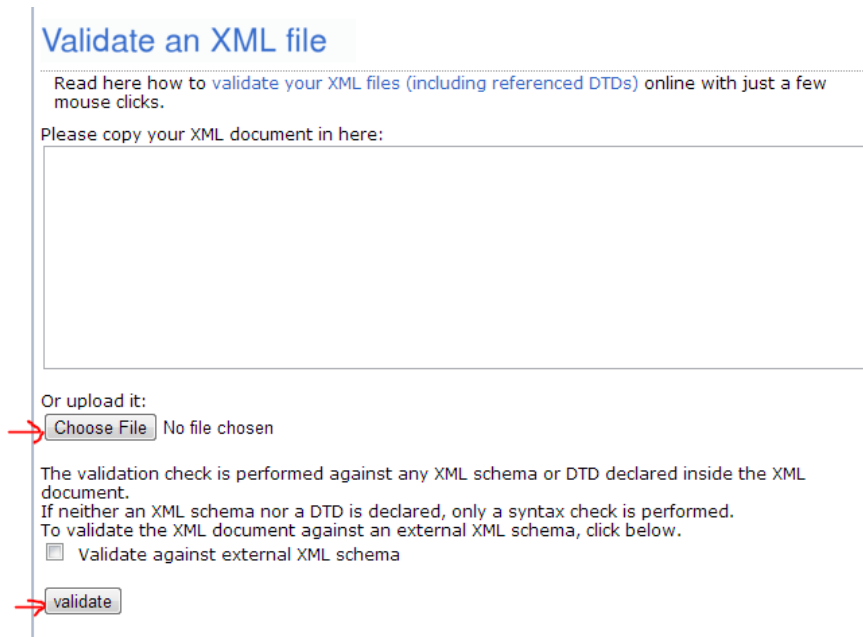
1. Prepare the XML file for validation. The first line should read:

```
<?xml version="1.0"?><!DOCTYPE HistPlugin SYSTEM "HistPlugin.dtd">
```

Note: This is only for validation purposes! Omit the DOCTYPE node completely in production files.

2. Navigate to www.xmlvalidation.com.
3. Click on **Choose File** to upload your XML file. Or, copy and paste the XML from your file into the space reserved for it.

Next, click the **validate** button.



The screenshot shows the 'Validate an XML file' page. At the top, there's a heading 'Validate an XML file' and a sub-header 'Read here how to validate your XML files (including referenced DTDs) online with just a few mouse clicks.' Below this is a text area with the prompt 'Please copy your XML document in here:'. Underneath the text area is the section 'Or upload it:' which contains a 'Choose File' button (highlighted with a red arrow) and the text 'No file chosen'. Below this is a paragraph explaining the validation process: 'The validation check is performed against any XML schema or DTD declared inside the XML document. If neither an XML schema nor a DTD is declared, only a syntax check is performed. To validate the XML document against an external XML schema, click below.' This is followed by a checkbox labeled 'Validate against external XML schema'. At the bottom, there is a 'validate' button (also highlighted with a red arrow).

4. A new window appears to inform you the DTD is required. Click on **Choose File** to upload HistPlugin.dtd. Or, copy and paste the contents of HistPlugin.dtd into the space reserved for it. Click on **continue validation** to start the validation.

Validate an XML file

Read here how to [validate your XML files \(including referenced DTDs\)](#) online with just a few mouse clicks.

The file HistPlugin.dtd is being referenced. Please copy it in here, so that the validation can continue:

Or upload it:

→ No file chosen

→

The following files have been uploaded so far:

XML document:

Click on any file name if you want to edit the file.

5. If you get a window that says “**No errors were found**”, your XML file is correct.

XML Validation

Validate an XML file

Read here how to [validate your XML files \(including referenced DTDs\)](#) online with just a few mouse clicks.

No errors were found

The following files have been uploaded so far:

XML document:

HistPlugin.dtd

Click on any file name if you want to edit the file.

6. If there are errors, they will be displayed. Follow the instructions on the screen to understand them. If it seems every line has an error, it is likely something is wrong at the beginning of your XML. Some basic things to check include:
 - a. Are the header and DOCTYPE correct? See step 1.
 - b. Did you forget the root node, <HistPlugin>?
 - c. Did you forget to include the end tag for any node?
7. To fix the problem, you can make changes to the XML or DTD. Scroll to the bottom of the document, and click on **XML Document:** or **HistPlugin.dtd**. When you're done editing, click the appropriate button to validate again.